

Document Summary TASK-FC7ECFE4

Analysis & Generated Response

1. Project Overview

- **Project Title:** Sovereign On-Premise Agentic AI Workbench using Open-Weight Multimodal LLMs for Confidential Industrial Work
- **Problem Statement ID:** SIH26117
- **Team Name/Author:** TEAM ADRESTIA
- **Theme:** Smart Automation

2. Context & Problem Statement

The problem statement identifies a critical challenge in the current context, which is to address the operational difficulty of securing confidential data within an on-premise environment while ensuring that AI-driven solutions are secure and reliable.

3. Technical Solution & Architecture

The technical solution addresses the core challenges by leveraging open-source technologies such as Qwen (Qwen), Meta Llama, Mistral, Gemma, PaddleOCR, OCR + VLM pipelines, LangChain, LlamaIndex, Haystack, FAISS, and Qdrant to create a sovereign AI workbench. The architecture is modularized with the following components:

- **Model Router:** Local models are selected per task for efficient execution.
- **Task-Aware Model Router:** Ensures that each model performs best according to its specific tasks.
- **Multi-step Work:** Automated workflows involving multiple steps, including data scanning, multimodal processing, and verification of actions.

4. Key Innovations & Security

The key innovations include:

- **Open-Weight AI:** Utilizing open-source models with locally deployed components that can evolve without redesigning the core.
- **Local OCR + Vision Models:** Leveraging local OCR and vision capabilities to improve accuracy in data processing tasks.
- **Private RAG (Rag-Aggregation):** Aggregating private source documents for a more secure and controlled execution environment.

5. Practical Impact & Deliverables

The project aims to:

- Ensure the security of confidential data within an on-premise AI workbench, reducing risks associated with cloud-based solutions.
- Provide a sovereign AI platform that can be deployed in various industries without compromising on control or scalability.
- Offer tools and processes for automated verification and approval workflows.

6. Feasibility & Viability

The feasibility of the project is supported by:

- **Open-Weight:** No vendor lock-in, allowing for controlled execution with minimal redesigning of core components.
- **Local Knowledge:** Leveraging proven open technologies to ensure secure and reliable AI-driven solutions within a sovereign environment.

7. Impact on Industry

- *Industry

Grounding & Evidence Sources

- **final ppt sih adrestia.pptx** (Referenced): TITLE PAGE SMART INDIA HACKATHON 2026 Problem Statement ID – SIH26117 Problem Statement Title- “Sovereign On-Premise Agentic AI Workbench u